

Harsh Shukla

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EDUCATION

McMaster University

Bachelor of Engineering in Software Engineering (Co-op), GPA: 3.9/4.0

Hamilton, ON

Sep 2023 – Apr 2028 (Expected)

EXPERIENCE

Software Developer

Intact Financial Corporation

Sept 2025 – Present

Toronto, Canada

- Engineered core components of the Enterprise Content Management (ECM) service using **Java Spring Boot**, optimizing document retrieval speeds by **25%** for internal stakeholders.
- Architected a cloud-native Claims Management System using **AWS Lambda**, **API Gateway**, and **DynamoDB**, migrating legacy data to reduce query latency by **45%** and operational costs by **30%**.
- Resolved **15+** critical performance bottlenecks by implementing **MongoDB** index optimizations and rigorous **JUnit** testing, increasing system scalability by **30%**.

Full Stack Software Engineering Intern

Laurette Software

Apr 2025 – Aug 2025

Remote

- Deployed a scalable school directory on **AWS & Vercel**, serving **100+** active users by architecting a normalized **PostgreSQL** schema for complex relationship mapping.
- Secured user data by implementing **Auth0** authentication with **JWT** strategy and Role-Based Access Control (RBAC), maintaining **99.9%** system availability.
- Modernized the frontend architecture with **Material-UI**, reducing administrative data entry time by **60%** through intuitive UI/UX design.

Software Developer, Exoskeleton Team

McMaster University

Dec 2024 – Aug 2025

Hamilton, ON

- Developed a high-frequency data pipeline for 10+ inertial sensors using **C++** and **ROS**, slashing latency by **60%** to enable real-time gait control.
- Implemented an Extended Kalman Filter (EKF) for multi-sensor fusion, improving positional tracking accuracy by **35%** during dynamic movement.
- Deployed AI-driven **gait prediction models** in **Python**, achieving **92%** accuracy in stride phase detection for adaptive robotic assistance.

PROJECTS

CyberSecurity System

Nov 2024

- Designed a comprehensive system integrating **facial recognition**, a password manager, & file **encryption** capabilities using real-time video & symmetric key encryption.
- Leveraged technologies including **Python** (Flask, OpenCV, Cryptography) & **SQL** to ensure robust functionality with less than **2ms** response time.
- Achieved 2nd place among 30+ teams, recognized for technical innovation & practical execution.

Astra: Multi-Agent Educational Tool

Dec 2025

- Engineered a multi-agent orchestration layer using **Python** & **LangGraph** to provide students with a 24/7 Socratic Tutor that autonomously adapts teaching styles based on identified knowledge gaps.
- Developed an automated "Insights Agent" for educators that analyzes student-tutor interactions to generate real-time difficulty heatmaps, reducing manual assessment overhead by **50%**.
- Architected a **RAG** pipeline using **Pinecone** & **OpenAI API** to ground agent responses in specific course syllabi, achieving **98%** factual accuracy while enabling teachers to adjust pedagogical parameters via a **React** dashboard.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frameworks/Tools: React, Node.js, Docker, Kubernetes, Firebase, Azure, ROS, Git, GitHub Actions, Jenkins, Postman, UML, Auth0, JWT

Concepts: Data Structures & Algorithms, OOP, RESTful APIs, Microservices, Agile/Scrum, CI/CD